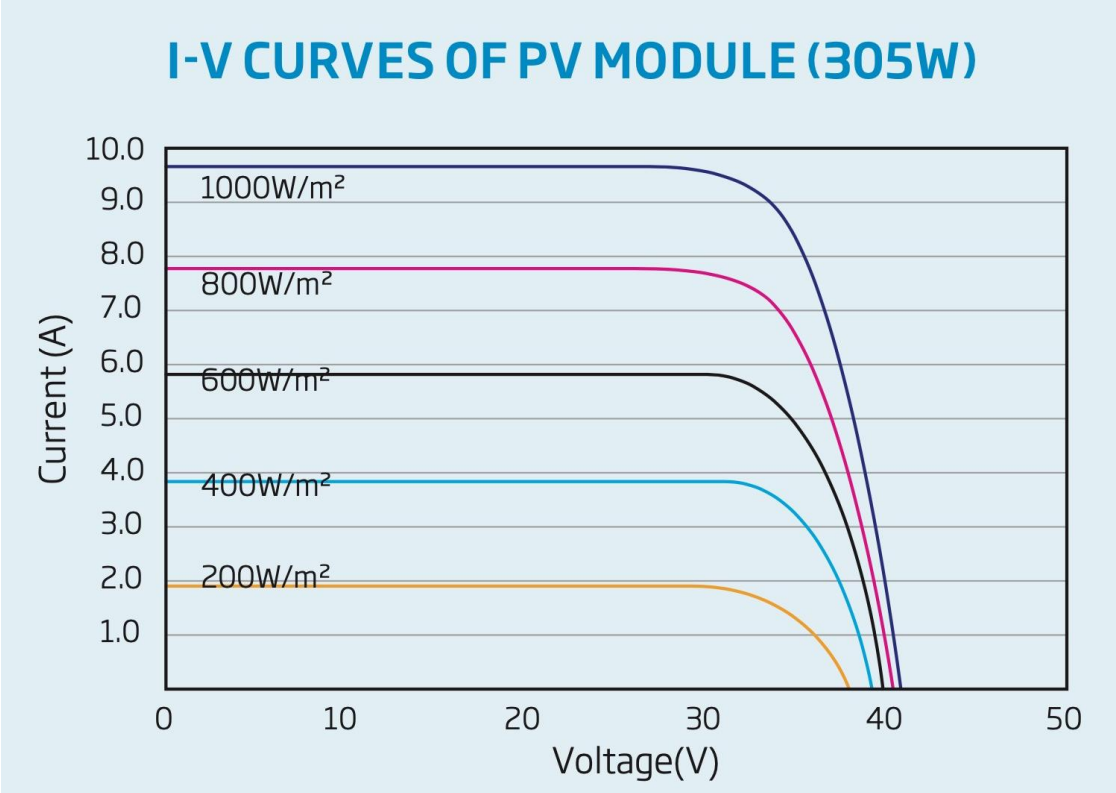
Experiment 5

Photovoltaic cells

1) Draw the IV curve characteristic and power curve for a PV module showing the main important parameters.



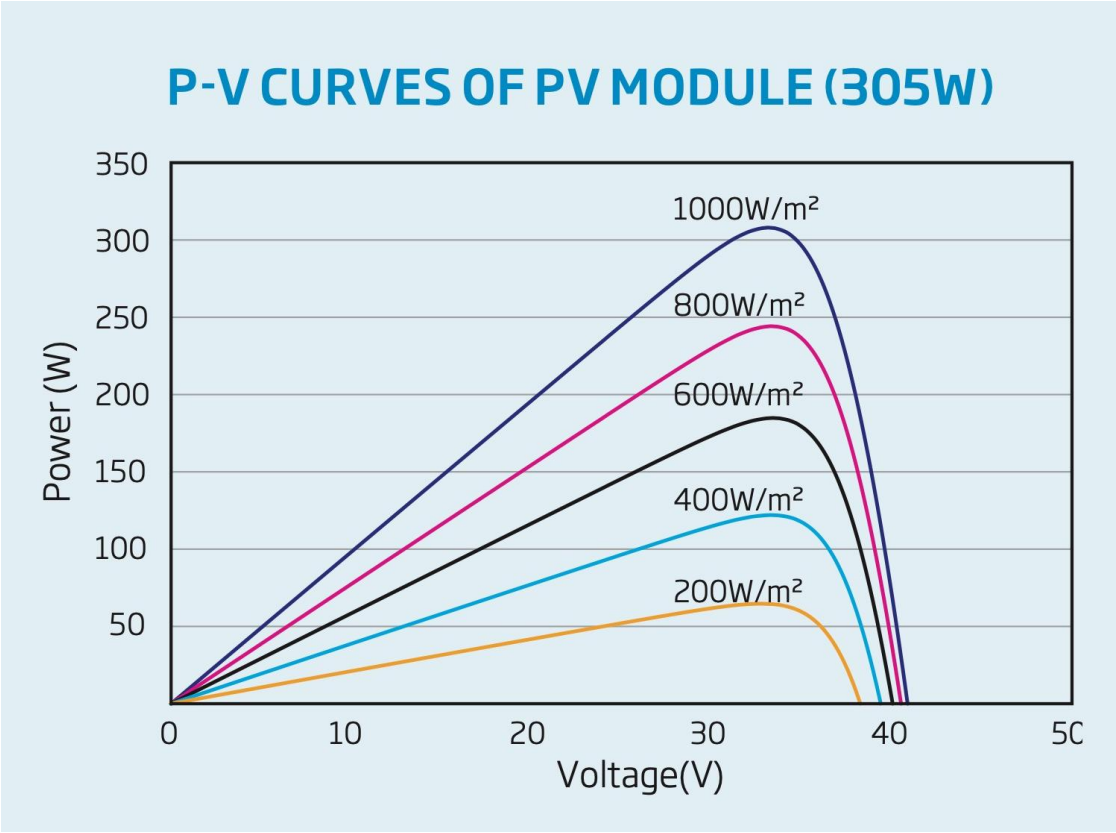
2) Explain the operation of the P&O algorithm showing its main disadvantages.

The Perturb and Observe (P&O) algorithm is a common technique used in Maximum Power Point Tracking (MPPT) for photovoltaic systems.
The algorithm perturbs (i.e., slightly adjusts) the operating voltage or current of the PV panel and observes the effect on power output.
- If the power increases, the algorithm continues to perturb in the same direction.
- If the power decreases, it reverses the direction of perturbation.

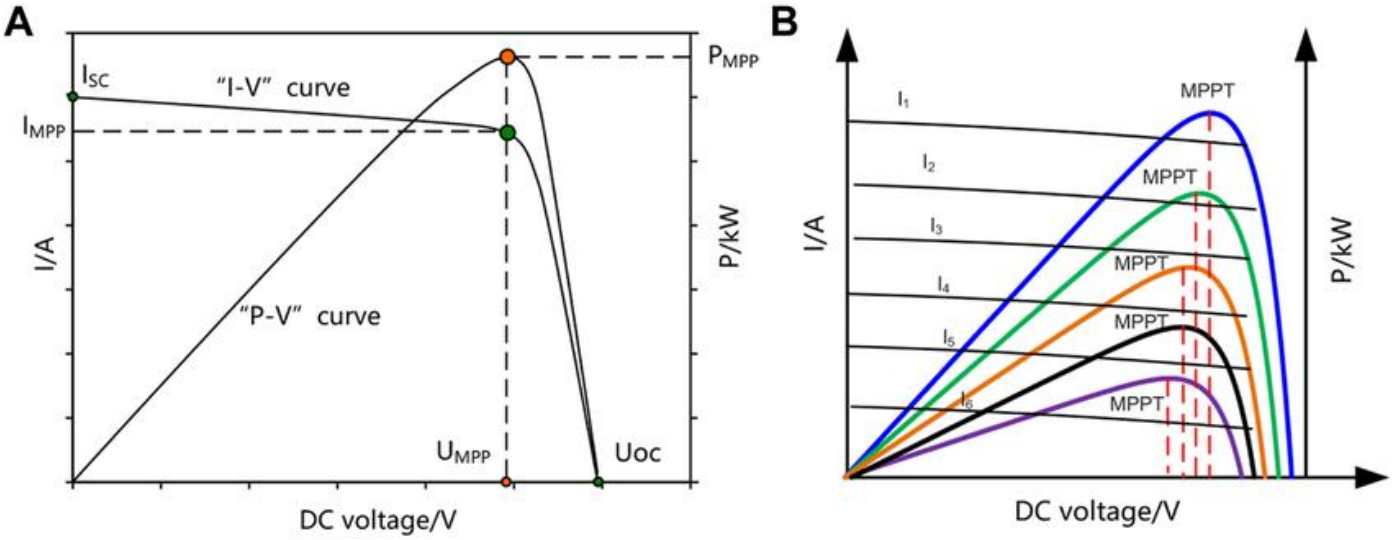
Main Disadvantages:
- Oscillations around the Maximum Power Point (MPP) due to continuous perturbation.
- Performance degradation under rapidly changing environmental conditions, such as irradiance and temperature variations.
- May misinterpret power changes due to irradiance shifts as MPP movement, leading to incorrect perturbation direction.

3) Using the Simulink model show the effect of:

 Change in Irradiance on the Power Output (Show the power curve for the base case & the power curve for another case with another irradiance)



 P&O MPPT (Show the power curve for the base case with P&O MPPT enabled & the power curve for another case with P&O MPPT disabled)



PV output voltage with and without MPPT control

